THE MODULE OF MAPS INTO A TORSION FREE MODULE IS TORSION FREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of the cards already published from this chapter, so that the batch selector treats them as one source instead of deferring them as a same-source clash.

Allowed standard prerequisites (editorial).

- Every module is a quotient of a direct sum of copies of the ring. used at proof line 5002, the presentation chosen for the arbitrary module M. Supplies the surjection whose dual inclusion puts Hom_R(M, N) inside a product of copies of N.
- A submodule of a torsion free module is torsion free, and torsion is tested elementwise. used at proof lines 5003-5004 together with the definition at source lines 4781-4792. Transports torsion freeness from the product of copies of N to Hom_R(M, N).
- Contravariance of Hom in its first variable. used at proof lines 5002-5003, standard for the Hom functor. Turns a surjection onto M into an inclusion of the Hom module into the product, which is the whole argument.

Source reference. Original lemma 0AUX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13665–13665. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let R be a domain. Let M, N be R-modules. If N is torsion free, so is* $\mathrm{Hom}_R(M, N)$.

Proof. Choose a surjection $\bigoplus_{i \in I} R \to M$. Then $\mathrm{Hom}_R(M, N) \subset \prod_{i \in I} N$. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.